

cannot be used without further information on the distribution of the errors. In principle, this problem can be tackled by estimating the multiplicative model using a consistent estimator, and then obtaining the appropriate weights estimating the skedastic function non-parametrically, as suggested by Delgado (1992) and Delgado and Kniesner (1997). However, this nonparametric generalized least squares estimator is rather cumbersome to implement, especially if the model has a large number of regressors. Moreover, the choice of the first round estimator is an open question as the NLS estimator may be a poor starting point due to its considerable inefficiency. Therefore, the nonparametric generalized least squares estimator is not appropriate to use as a work-horse for routine estimation of multiplicative models.¹⁰ Indeed, what is needed is an estimator that is consistent and reasonably efficient under a wide range of heteroskedasticity patterns and is also simple to implement.

A possible way of obtaining an estimator that is more efficient than the standard NLS without the need to use nonparametric regression is to follow McCullagh and Nelder (1989) and estimate the parameters of interest using a pseudo-maximum likelihood estimator based on some assumption on the functional form of $V[y_i|x]$.¹¹ Among the many possible specifications, the hypothesis that the conditional variance is proportional to the conditional mean is particularly appealing. Indeed, under this assumption $E[y_i|x] = \exp(x_i\beta) \propto V[y_i|x]$, and β can be estimated by solving the following set of first order conditions:

$$\sum_{i=1}^n \left[y_i - \exp(x_i\tilde{\beta}) \right] x_i = 0. \quad (9)$$

Comparing equations (8) and (9), it is clear that, unlike the NLS estimator, which is a PML estimator obtained assuming that $V[y_i|x]$ is constant, the PML estimator based on (9) gives the same weight to all observations, rather than emphasizing those for which

¹⁰A nonparametric generalized least squares estimator can also be used to estimate linear models in presence of heteroskedasticity of unknown form (Robinson, 1987). However, despite having been proposed more than 15 years ago, this estimator has never been adopted as a standard tool by empirical researchers, who generally prefer the simplicity of the inefficient OLS, with an appropriate covariance matrix.

¹¹See also Manning and Mullahy (2001). A related estimator is proposed by Papke and Wooldridge (1996) for the estimation of models for fractional data.

$\exp(x_i\beta)$ is large. This is because, under the assumption that $E[y_i|x] \propto V[y_i|x]$, all observations have the same information on the parameters of interest as the additional information on the curvature of the conditional mean coming from observations with large $\exp(x_i\beta)$ is offset by their larger variance. Of course, this estimator may not be optimal, but without further information on the pattern of heteroskedasticity, it seems natural to give the same weight to all observations.¹² Even if $E[y_i|x]$ is not proportional to $V[y_i|x]$, the PML estimator based on (9) is likely to be more efficient than the NLS estimator when the heteroskedasticity increases with the conditional mean.

The estimator defined by (9) is numerically equal to the Poisson pseudo-maximum likelihood (PPML) estimator which is often used for count data.¹³ The form of (9) makes clear that all that is needed for this estimator to be consistent is the correct specification of the conditional mean, i.e., $E[y_i|x] = \exp(x_i\beta)$. Therefore, the data do not have to be Poisson at all and, what is more important, y_i does not even have to be an integer, for the estimator based on the Poisson likelihood function to be consistent. This is the well-known pseudo-maximum likelihood result first noted by Gourieroux, Monfort and Trognon (1984).

The implementation of the PPML estimator is straightforward since there are standard econometric programs with commands that permit the estimation of Poisson regression, even when the dependent variables are not integers. Because the assumption $V[y_i|x] \propto E[y_i|x]$ is unlikely to hold, this estimator does not fully account for the heteroskedasticity in the model and all inference has to be based on an Eicker-White (Eicker, 1963; and White, 1980) robust covariance matrix estimator.

Of course, if it was known that $V[y_i|x]$ is a function of higher powers of $E[y_i|x]$, a more efficient estimator could be obtained down-weighting even more the observations with large conditional mean. An example of such estimator is the gamma pseudo-maximum likelihood estimator studied by Manning and Mullahy (2001) which, like the log-linearized

¹²The same strategy is implicitly used by Papke and Wooldridge (1996) in their pseudo-maximum estimator for fractional data models.

¹³See Cameron and Trivedi (1998) and Winkelmann (2003) for more details on the Poisson regression and on more general models for count data.

model, assumes that $V[y_i|x]$ is proportional to $E[y_i|x]^2$. The first order conditions for the gamma PML estimator are given by

$$\sum_{i=1}^n \left[y_i - \exp\left(x_i\check{\beta}\right) \right] \exp\left(-x_i\check{\beta}\right) x_i = 0.$$

In the case of trade data, however, this estimator may have an important drawback. Trade data for larger countries (as gauged by GDP per capita) tend to be of higher quality (see Frankel and Wei, 1993, and Frankel 1997); hence, models assuming that $V[y_i|x]$ is a function of higher powers of $E[y_i|x]$ might give excessive weight to the observations that are more prone to measurement errors.¹⁴ Therefore, the Poisson regression emerges as a reasonable compromise, giving less weight to the observations with larger variance than the standard NLS estimator, without giving too much weight to observations more prone to contamination by measurement error and less informative about the curvature of $E[y_i|x]$.

3.2 Testing

In this section we consider tests for the particular pattern of heteroskedasticity assumed by PML estimators, focusing on the PPML estimator. Although PML estimators are consistent even when the variance function is misspecified, the researcher can use these tests to check if a different PML estimator would be more appropriate and to decide whether or not the use of a nonparametric estimator of the variance is warranted.

Manning and Mullahy (2001) suggested that if

$$V[y_i|x] = \lambda_0 E[y_i|x]^{\lambda_1}, \quad (10)$$

the choice of the appropriate PML estimator can be based on a Park-type regression (Park, 1966). Their approach is based on the idea that if (10) holds and an initial

¹⁴Frankel and Wei (1993) and Frankel (1997) suggest that larger countries should be given more weight in the estimation of gravity equations. This would be appropriate if the errors in the model were just the result of measurement errors in the dependent variable. However, if it is accepted that the gravity equation does not hold exactly, measurement errors account for only part of the dispersion of trade data around the gravity equation.

consistent estimate of $E[y_i|x]$ is available, then λ_1 can be consistently estimated using an appropriate auxiliary regression. Specifically, following Park (1966), Manning and Mullahy (2001) suggest that λ_1 can be estimated using the auxiliary model

$$\ln(y_i - \check{y}_i)^2 = \ln(\lambda_0) + \lambda_1 \ln(\check{y}_i) + v_i, \quad (11)$$

where $\check{y}_i$ denotes the estimated value of $E[y_i|x]$. Unfortunately, as the discussion in the previous sections should have made clear, this approach based on the log-linearization of (10) is valid only under very restrictive conditions on the conditional distribution of y_i . However, it is easy to see that this procedure is valid when the constant-elasticity model can be consistently estimated in the log-linear form. Therefore, using (11) a test for $H_0 : \lambda_1 = 2$ based on a non-robust covariance estimator provides a check for the adequacy of the estimator based on the log-linear model.

A more robust alternative, which is mentioned by Manning and Mullahy (2001) in a footnote, is to estimate λ_1 from

$$(y_i - \check{y}_i)^2 = \lambda_0 (\check{y}_i)^{\lambda_1} + \xi_i, \quad (12)$$

using an appropriate PML estimator. The approach based on (12) is asymptotically valid and inference about λ_1 can be based on the usual Eicker-White robust covariance matrix estimator. For example, the hypothesis that $V[y_i|x]$ is proportional to $E[y_i|x]$ is accepted if the appropriate confidence interval for λ_1 contains 1. However, if the purpose is to test the adequacy of a particular value of λ_1 , a slightly simpler method based on the Gauss-Newton regression (see Davidson and MacKinnon, 1993) is available.

Specifically, to check the adequacy of the PPML for which $\lambda_1 = 1$ and $\check{y}_i = \exp(x_i\tilde{\beta})$, (12) can be expanded in a Taylor series around $\lambda_1 = 1$, leading to

$$(y_i - \check{y}_i)^2 = \lambda_0 \check{y}_i + \lambda_0 (\lambda_1 - 1) \ln(\check{y}_i) \check{y}_i + \xi_i.$$

Now, the hypothesis that $V[y_i|x] \propto E[y_i|x]$ can be tested against (10) simply by checking the significance of the parameter $\lambda_0(\lambda_1 - 1)$. Because the error term ξ_i is unlikely to be homoskedastic, the estimation of the Gauss-Newton regression should be performed using